

FUNCTIONALS Anonymous authors Paper under double-blind review	# ABSTRACT The accuracy of density functional theory hinges on the approximation of non-local contributions to the exchange-correlation (XC) functional. To date, machine-learned and human-designed approximations suffer from insufficient accuracy, limited scalability, or dependence on costly reference data. To address these issues, we introduce Equivariant Graph Exchange Correlation (EG-XC), a novel non-local XC functional based on equivariant graph neural networks (GNNs). Where previous works relied on semi-local functionals or fixed-size descriptors of the density, we compress the electron density into an $SO(3)$-equivariant nuclei-centered point cloud representation for efficient non-local atomic-range interactions. By applying an equivariant GNN on this point cloud, we capture molecular-range interactions in a scalable and accurate manner. To train EG-XC, we differentiate through a self-consistent field solver requiring only energy targets. In our empirical evaluation, we find EG-XC accurately reconstructs 'gold-standard' CCSD(T) energies on MD17. On out-of-distribution conformations of 3BPA, EG-XC reduces the relative MAE by 33% to 50%. Remarkably, EG-XC excels in data efficiency and molecular size extrapolation on QM9, matching force fields trained on 5 times more and larger molecules. On identical training sets, EG-XC yields on average 51% lower MAEs.	# 1 INTRODUCTION Kohn-Sham Density Functional Theory (KS-DFT) is the backbone of computational material and drug discovery (Jones, 2015). It is a quantum mechanical method to approximate the ground state energy of an N-electron system by finding the electron density ρ that minimizes the energy functional $E[\rho]$. This functional maps electron densities to energies and is composed of $E[\rho] = T[\rho] + V_{\text{ext}}[\rho] + J[\rho] + E_{\text{xc}}[\rho]$ $T[\rho]$ is the kinetic energy functional, $V_{\text{ext}}[\rho]$ is the external potential due to positively charged nuclei, $J[\rho]$ is the Coulomb energy between electrons, and $E_{\text{xc}}[\rho]$ is the exchange-correlation (XC) functional (Cramer, 2004). While $T[\rho]$ and $V_{\text{ext}}[\rho]$ permit analytical computations, the exact form of $E_{\text{xc}}[\rho]$ remains unknown, and its approximation frequently dominates DFT's error (Kim et al., 2013).	# 2 BACKGROUND Notation. To denote functionals, i.e., functions mapping from functions to scalars, we write $\mathcal{F}[\cdot]$ where $\mathcal{F}$ is the functional and $\mathbf{f}$ the function. To denote superscripts in brackets $\mathbf{f}^{(i)}$ to indicate sequences and regular superscripts of bold vectors, e.g., $\mathbf{f}^{(i)}$, to indicate the i-th irreducible representation of the $SO(3)$ group. We generate $\mathbf{r}$ to denote points in the 3D Euclidean space. A notable exception is the finite set of nuclei positions $\{\mathbf{R}_1, \dots, \mathbf{R}_N\}$ for which we use capital letters. The norm $\ \mathbf{r}\$ is given by $\ \mathbf{r}\ = \sqrt{\mathbf{r} \cdot \mathbf{r}}$. For directions, i.e., unit length vectors, we use $\hat{\mathbf{r}} = \mathbf{r} / \ \mathbf{r}\$. Kohn-Sham density functional theory is the foundation of our work. Here, we provide a very brief introduction. For more details, we refer the reader to Appendix A or Lehtola et al. (2020). In KS-DFT, the electron density ρ is represented by a set of orthogonal orbitals $\{\phi_i\}$ that are defined as linear combinations of a basis set of atomic orbitals $\{\chi_\mu\}$ to	# 3 RELATED WORK Learnable density functional approximations (DFAs) of the XC functional have a long history. In their seminal paper, Kohn & Sham (1965) proposed an ϵ-functional fitted to a reference calculation. In this first parameterization ϵ only depends on $\rho(r) = \rho(\mathbf{r})$. To improve accuracy, the density features $\mathbf{f}$ have been successively refined by including gradients and physical quantities such as the kinetic energy density $\tau(r) = \frac{1}{2} \sum_i \nabla \phi_i(\mathbf{r}) ^2$. The resulting semi-local functionals are also known as meta-GGAs with $g(r) = g(\mathbf{r})$. Many ML functionals use this parameterization with ϵ frequently being a product of an MLP with a classical functional (Nagai et al., 2022; Kulik et al., 2022; Zhang et al., 2023). An appealing aspect of learning semi-local energy densities rather than ϵ directly is the known implementation of physical constraints (Sun et al., 2015; Kaplan et al., 2023), which inspired works by Dick & Fernandez-Serra (2021); Nagai et al. (2022). While augmenting g with hand-crafted non-local features improves the accuracy, this comes at the cost of exact constraints (Nagai et al., 2020; Kirkpatrick et al., 2021). Similarly, such physical biases are absent in kernel methods Margraf & Neuter (2021); Byström & Kozinsky (2022). Additionally, these methods require reference densities that are rarely available (Szabo & Ostlund, 2012). Alternative grid-based GNNs converge slowly in the number of grid points compared to specialized spherical integration grids (Zhou et al., 2019; Treutler & Ahlrichs, 1995). Lastly, machine-learned functionals have been used in other DFT contexts as well, e.g., for energy corrections on SCF-converged energies (Chen & Yang, 2024) or to model the kinetic energy functional in orbital-free DFT (Snyder et al., 2012; Mi et al., 2023; Zhang et al., 2024). Here, we propose to extend $\mathbf{f}$ to the so-called density matrix. Given the representation in a finite basis set, one can compute the kinetic energy T, external potential V_{ext}, and electron-electron repulsion J.																		
# 4 EQUIVARIANT GRAPH EXCHANGE CORRELATION With Equivariant Graph Exchange Correlation (EG-XC), we propose an efficient non-local approximation to the unknown exchange-correlation functional E_{xc} that maps electron densities (positive integrals) to scalar energies. To the end, EG-XC consists of four components as illustrated in Figure 1: (1) Nuclei-centered equivariant embeddings $\mathbf{f}_{\text{nc}}$ compress the electron density ρ into a finite point cloud representation by equivariantly integrating the density around the nuclei enabling atomic-range interactions. (2) Equivariant message passing on this point cloud includes molecular-range information in $\mathbf{f}_{\text{nc}}$. Based on the point cloud embeddings, we define E_{xc} as the sum of two readouts: (3) A non-local reweighted meta-GGA based on our non-local feature density $\mathbf{f}_{\text{nc}}$ and (4) a graph readout the invariant nuclei-centered features $\mathbf{f}_{\text{nc}}$: $E_{\text{xc}}[\rho] = \sum_{\mathbf{r}} \left(\sum_{\mathbf{r}'} \mathbf{f}_{\text{nc}}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{f}_{\text{nc}}(\mathbf{r}', \mathbf{r}) \right) + \sum_{\mathbf{r}} \mathbf{f}_{\text{nc}}(\mathbf{r}) \cdot \mathbf{f}_{\text{nc}}(\mathbf{r})$	# 5 EXPERIMENTS In the following, we compare EG-XC to various alternative methods of learning potential energy surfaces across several settings. In particular, we focus on the following tasks: interpolating accurate energy surfaces, extrapolation to unseen conformations, and extrapolation to larger molecules. Additionally, we present an ablation study on EG-XC's components. For a runtime complexity analysis, we refer the reader to Appendix M and for runtime measurements to Appendix N. Methods. To accurately position EG-XC, we compare to methods of varying computational costs. In particular, we evaluate force fields, Δ-ML, i.e., combining force fields with KS-DFT calculations, and a learnable XC-functional (Dick & Fernandez-Serra, 2021). While force fields are orders of Table 1: Test set MAE on the CCSD(T) MD17 dataset in mHa. (best, second) <table><tr><th>Method</th><th>MAE (mHa)</th></tr><tr><td>EG-XC</td><td>0.01</td></tr><tr><td>KS-DFT</td><td>0.02</td></tr><tr><td>SchNet</td><td>0.03</td></tr><tr><td>PaiNN</td><td>0.04</td></tr><tr><td>NequIP</td><td>0.05</td></tr><tr><td>Batzner et al.</td><td>0.06</td></tr><tr><td>Dick & Fernandez-Serra</td><td>0.07</td></tr><tr><td>Delta-ML</td><td>0.08</td></tr><tr><td>Force fields</td><td>0.09</td></tr></table>	Method	MAE (mHa)	EG-XC	0.01	KS-DFT	0.02	SchNet	0.03	PaiNN	0.04	NequIP	0.05	Batzner et al.	0.06	Dick & Fernandez-Serra	0.07	Delta-ML	0.08	Force fields	0.09	# 6 DISCUSSION We tackled the problem of learning efficient non-local exchange-correlation functionals to enhance the accuracy of density functional theory. To this end, we have presented Equivariant Graph Exchange Correlation (EG-XC), an equivariant graph neural network approach. We have introduced a basis-independent reduction of the electron density into a finite point cloud representation, enabling graph neural networks to capture molecular-range information. Based on this point cloud embedding, we have parameterized a non-local feature density used for the reweighting of a semi-local exchange-correlation energy density. Unlike other learnable non-local functionals, EG-XC can be trained by differentiating through the SCF solver such that the training requires only energy targets. In our empirical evaluation, EG-XC improves upon previous machine learning-based methods of modeling potential energy surfaces. On MD17, EG-XC accurately reconstructs 'gold-standard' potential energy surfaces, namely CCSD(T), within the DFT framework. On the 3BPA, EG-XC reduces errors by 33% to 50% compared to the best-performing baseline. On QM9, EG-XC demonstrates remarkable data efficiency, achieving similar accuracies with 5 times less data or up to 33% lower error than the same amount of data compared to the best baseline. Finally, EG-XC extrapolates well to unseen conformations and larger molecules on 3BPA and QM9, respectively. Overall, these results strongly underline the data efficiency of learning exchange-correlation functionals compared to machine-learned force fields. We hypothesize that a significant portion of EG-XC's generalization is thanks to including the physically constrained semi-local model that captures most of the exchange-correlation energy and biases EG-XC to approximately fulfill the same constraints. On top of these, our non-local contributions transfer the accuracies of graph neural network-based force fields to functionals while maintaining a similar data efficiency.
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